



Trade Variability During Global Disruption



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ABSTRACT

Global disruptions such as the COVID-19 pandemic introduce sudden shocks and heightened volatility in international trade, challenging traditional time-series models that assume stable trends and relationships. Understanding whether these shocks represent temporary deviations or persistent structural changes is critical for policymakers, supply-chain planners, and economic forecasters. This study investigates how trade variability evolves before, during, and after major global disruptions, with a focus on identifying structural instability that conventional regression approaches may overlook.

Using official trade data from the [United Nations Comtrade database](#), this study analyzes bimonthly electrical machinery import data from 2017 to 2023 for six countries: the United States, Mexico, Canada, France, Japan, and China. The dataset consists of 505 observations and includes trade value (CIF and/or FOB values) and detailed time identifiers. This focus on multi-country panels allows for cross-country comparison while maintaining analytical clarity within a targeted commodity category.

We implement a layered time-series framework combining anomaly detection (Z-score analysis), baseline time-series regression, change detection using CUSUM, and regime-based modeling through piecewise regression. Rather than relying solely on prediction accuracy metrics such as RMSE or MAE, the evaluation emphasizes the timing, magnitude, and persistence of structural disruptions. This approach enables us to distinguish short-term volatility from sustained regime shifts and to assess whether disruption effects are transitory or long-lasting.

Preliminary findings indicate a synchronized decline in trade during 2020 across all countries, followed by heterogeneous recovery patterns. Structural break tests suggest that disruption effects cluster around identifiable breakpoints, with varying persistence across economies. The results indicate that the COVID-19 shock generated a high-intensity, but temporally bounded disruption rather than a permanent regime shift in global trade dynamics.

More broadly, the proposed framework provides an interpretable and scalable methodology for detecting disruption timing, evaluating persistence, and comparing structural instability across countries. This approach can be extended to other commodities, additional economies, or future global disruptions. Implementing the analysis across both Python and R also improves transparency and reproducibility, ensuring that detected structural signals are not artifacts of a single computational environment.

INTRODUCTION

Project Motivation & Background

International trade plays a central role in global economic stability, supply chain coordination, and national growth. Over the past decade, trade systems have become increasingly interconnected, with production processes fragmented across countries and dependent on just-in-time logistics. While this improves efficiency during stable periods, it also increases vulnerability to systemic shocks. Events such as the COVID-19 pandemic exposed structural fragilities in global supply chains, leading to abrupt trade contractions, delayed shipments, and significant volatility across countries and sectors.

Traditional economic and time-series models often assume stable relationships over time, capturing gradual trends and seasonal fluctuations. However, during major disruptions, these assumptions may no longer hold. Sudden policy interventions, border closures, demand collapses, and supply constraints can generate non-linear behavior and structural breaks that are not well captured by standard regression or trend-based models. As a result, there is growing interest in approaches that focus on detecting structural instability rather than only predicting average outcomes.

This project is motivated by the need to better understand how international trade responds to large-scale disruptions and whether these responses are temporary shocks or longer-lasting regime shifts. By focusing on electrical machinery imports across six major economies (United States, Mexico, Canada, France, Japan, and China), this study provides a comparative framework to examine cross-country differences in trade disruption dynamics using data from the United Nations Comtrade database.

Business Justification: Why We Care

Understanding trade variability during global disruptions is not only an academic concern but also a strategic business priority. Multinational firms rely on stable cross-border trade flows to maintain inventory levels, production schedules, and distribution networks. When trade patterns shift unexpectedly, firms face increased costs, inventory imbalances, and revenue volatility. Early detection of structural instability can help businesses adjust sourcing strategies, diversify suppliers, and strengthen resilience planning.

From a policy perspective, governments need reliable indicators to determine whether trade contractions are temporary or reflect deeper economic restructuring. Misinterpreting a structural shift as a short-term fluctuation may delay necessary interventions, while overreacting to temporary volatility can distort long-term planning. By identifying the timing and persistence of disruptions, analytical frameworks such as the one proposed in this study can improve economic decision-making.

Global supply chains also amplify shocks across countries, meaning disruptions rarely remain localized. A shock in one major economy can propagate through production networks, logistics systems, and financial channels. As a result, businesses and policymakers need tools that move beyond simple averages and instead focus on

detecting instability, regime shifts, and volatility clustering. This project addresses that need by integrating anomaly detection, structural break testing, and regime-based modeling into a unified analytical framework.

Problem Statement

Global disruptions such as pandemics, geopolitical conflict, and supply chain interruptions can generate sharp volatility in international trade flows. Traditional trend-based analysis often struggles to distinguish routine cyclical fluctuations from true structural regime shifts in these environments.

The core problem this study addresses is:

How can we detect and compare structural trade instability across countries during major global disruptions, while distinguishing temporary shocks from persistent regime shifts?

Specifically, this study seeks to determine:

- Whether deviations observed during the COVID-19 period represent statistically significant structural breaks
- Whether these deviations are temporary shocks or persistent structural shifts
- Why countries exhibit different instability patterns (e.g., sharp one-month collapses vs. gradual structural drift)

While COVID-19 provides a clear empirical benchmark for validation, the framework developed in this study is designed to generalize to other disruptions such as geopolitical conflict, trade policy shocks, or large-scale supply chain interruptions.

To address this problem, we develop and validate a standardized cross-country modeling framework that integrates residual-based anomaly detection, threshold sensitivity testing, and cumulative persistence diagnostics.

Research Questions

The following are three significant research questions that our effort aims to address:

- **How does trade variability change before, during, and after major global disruptions? Specifically, do we observe measurable increases in volatility and abnormal deviations during disruption periods compared to baseline conditions?** Identifying these shifts is critical to understanding whether disruptions fundamentally alter trade dynamics.
- **Do global disruptions introduce persistent structural breaks in trade time series, or are deviations observed primarily as temporary deviations from long-term shocks?** Using structural break detection methods, such as CUSUM and piecewise regression, this study evaluates whether post-disruption trade behavior reflects a regime shift or reversion to trend. This distinction helps determine whether recovery represents stabilization within the existing framework or transition to a new equilibrium.

- **Do disruption responses differ systematically across countries in terms of timing, magnitude, and persistence?** Given differences in economic structure, supply chain exposure, and policy responses, countries may exhibit heterogeneous reactions in both the intensity and duration of trade shocks. By applying a standardized modeling framework across economies, this research compares whether structural instability manifests uniformly or varies according to transmission dynamics.

Initial Hypothesis

Based on preliminary exploratory analysis and prior literature on global supply chain disruptions, we propose several initial hypotheses.

First, we hypothesize that trade variability significantly increases during major global disruptions, with abnormal months clustering around identifiable breakpoints. These clusters are expected to reflect concentrated periods of instability rather than isolated random fluctuations.

Second, we hypothesize that traditional single-regime regression models will underestimate the magnitude and persistence of disruption effects. Structural and regime-based models such as piecewise regression and CUSUM are expected to better capture sustained deviations and shifts in trend behavior. In particular, we anticipate that structural break models will identify statistically meaningful change points corresponding to major global events.

Finally, we hypothesize that disruption effects will not be uniform across countries. While all six countries are expected to exhibit synchronized declines during peak disruption periods, the speed of recovery and degree of volatility are likely to differ. These differences may reflect varying levels of supply chain integration, industrial composition, and economic resilience.

Testing these hypotheses provides both methodological insight and economically meaningful interpretation of global trade instability.

METHODOLOGY & APPROACH



Figure 1 : Methodology & Approach

Our project utilized a structured methodology breaking down the objectives into phases and subtasks to address key components thoroughly. It is important to note that this is an iterative process that allows us to reframe steps and details as needed as evidence and knowledge become clear.

Approach:

- **Problem Definition**: This project examines how major global disruptions impact international trade variability and structural stability. Rather than focusing solely on forecasting trade levels, the objective is to detect and validate abnormal deviations and structural breaks across countries while distinguishing temporary volatility from persistent regime shifts.
- **Data Acquisition**: Monthly trade data (2017–2023) were sourced from the UN Comtrade database for six selected countries and a targeted commodity category. Time-consistent trade value variables were extracted and standardized to ensure cross-country comparability and reproducibility.
- **Data Exploratory Analysis**: Time-series visualization and summary statistics were used to identify volatility spikes, synchronized declines, and cross-country differences. Exploratory diagnostics informed baseline modeling specifications and highlighted potential disruption windows requiring formal structural testing.
- **Data Preparation**: The dataset was cleaned by removing irrelevant or constant variables and addressing missing values. Time-based features and standardized variables were constructed to support anomaly detection and cross-country structural comparison.
- **Modeling**: A layered framework was implemented, combining residual Z-score anomaly detection, baseline time-series regression (trend + seasonality), CUSUM diagnostics, and piecewise regression analysis. These models capture the timing and magnitude of abnormal deviations as well as their cumulative persistence, enabling clear differentiation between short-term shocks and regime-dependent structural change. This multi-method approach allows structural instability to be evaluated from multiple statistical perspectives, reducing the risk that detected disruption signals arise from assumptions specific to any single model. In addition to improving detection accuracy, the layered modeling structure enhances interpretability. Each method captures a different dimension of disruption behavior:
 - Anomaly detection identifies extreme short-term deviations
 - Piecewise regression models structural shifts in trend behavior
 - CUSUM diagnostics evaluate whether these deviations accumulate into persistent instability over time.
 - ARIMA modeling captures underlying dynamics of the series by incorporating past errors and autocorrelations. It also allows deviations from expected trade patterns to be evaluated.

By integrating these complementary perspectives, the framework provides a more comprehensive assessment of trade disruption dynamics than any single technique used in isolation.

- **Evaluation/Findings**: Models were evaluated using RMSE and MAE as reference metrics, with emphasis placed on disruption, timing, and persistence. Performance was also assessed using structural validation by using:
 - Timing consistency across countries,

- ## OVERVIEW OF DATA

We utilize the [United Nations Comtrade database](#) dataset maintained by the United Nations. Electrical machinery (HS85) was selected because it represents a core intermediate goods category within global manufacturing supply chains and therefore provides a useful benchmark for evaluating disruption transmission across economies.

The original dataset contained 505 observations and 47 variables, including trade values, country identifiers, commodity classifications, valuation metrics, and administrative reporting fields.

After cleaning and filtering, the **Figure 2 : Raw Data after Download** preserves key economic measures while removing redundant metadata and system-generated fields that do not contribute to modeling.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	refPeriodId	refYear	refMonth	period	reporterCode	reporterISO	reporterDesc	classificationCode	cmdDesc	isNetWgtEstimated	cifvalue	fobvalue	primaryValue	legacy Estimation Flag
1	20170101	2017	1	201701	124	CAN	Canada	H5	Electrical machinery and equipment a	FALSE	\$-	\$3,134,950,123.12	\$3,134,950,123.12	0
2	20170101	2017	1	201701	156	CHN	China	H5	Electrical machinery and equipment a	TRUE	\$28,636,636,234.00	\$-	\$28,636,636,234.00	4
3	20170101	2017	1	201701	251	FRA	France	H5	Electrical machinery and equipment a	TRUE	\$4,524,332,681.01	\$-	\$4,524,332,681.01	4
4	20170101	2017	1	201701	392	JPN	Japan	H5	Electrical machinery and equipment a	TRUE	\$8,717,372,419.27	\$-	\$8,717,372,419.27	4
5	20170101	2017	1	201701	484	MEX	Mexico	H4	Electrical machinery and equipment a	FALSE	\$-	\$6,081,945,379.00	\$6,081,945,379.00	0
6	20170101	2017	1	201701	842	USA	USA	H5	Electrical machinery and equipment a	TRUE	\$27,421,068,433.00	\$27,017,857,840.00	\$27,421,068,433.00	4
7	20170201	2017	2	201702	124	CAN	Canada	H5	Electrical machinery and equipment a	FALSE	\$-	\$3,057,570,518.75	\$3,057,570,518.75	0
8	20170201	2017	2	201702	156	CHN	China	H5	Electrical machinery and equipment a	TRUE	\$30,515,256,116.00	\$-	\$30,515,256,116.00	4
9	20170201	2017	2	201702	251	FRA	France	H5	Electrical machinery and equipment a	TRUE	\$3,857,656,209.45	\$-	\$3,857,656,209.45	4
10	20170201	2017	2	201702	392	JPN	Japan	H5	Electrical machinery and equipment a	TRUE	\$6,530,855,992.06	\$-	\$6,530,855,992.06	4
11	20170201	2017	2	201702	484	MEX	Mexico	H4	Electrical machinery and equipment a	FALSE	\$-	\$5,993,017,127.00	\$5,993,017,127.00	0
12	20170201	2017	2	201702	842	USA	USA	H5	Electrical machinery and equipment a	TRUE	\$22,901,663,960.00	\$22,564,845,866.00	\$22,901,663,960.00	4
13	20170301	2017	3	201703	124	CAN	Canada	H5	Electrical machinery and equipment a	FALSE	\$-	\$3,543,121,009.85	\$3,543,121,009.85	0
14	20170301	2017	3	201703	156	CHN	China	H5	Electrical machinery and equipment a	TRUE	\$33,988,876,164.00	\$-	\$33,988,876,164.00	4
15	20170301	2017	3	201703	251	FRA	France	H5	Electrical machinery and equipment a	TRUE	\$4,325,704,346.37	\$-	\$4,325,704,346.37	4
16	20170301	2017	3	201703	392	JPN	Japan	H5	Electrical machinery and equipment a	TRUE	\$7,967,886,451.70	\$-	\$7,967,886,451.70	4
17	20170301	2017	3	201703	484	MEX	Mexico	H4	Electrical machinery and equipment a	FALSE	\$-	\$7,242,160,543.00	\$7,242,160,543.00	0
18	20170301	2017	3	201703	842	USA	USA	H5	Electrical machinery and equipment a	TRUE	\$28,721,088,346.00	\$28,312,766,510.00	\$28,721,088,346.00	4
19	20170401	2017	4	201704	124	CAN	Canada	H5	Electrical machinery and equipment a	FALSE	\$-	\$3,297,111,761.50	\$3,297,111,761.50	0
20	20170401	2017	4	201704	156	CHN	China	H5	Electrical machinery and equipment a	TRUE	\$32,373,924,665.00	\$-	\$32,373,924,665.00	4
21	20170401	2017	4	201704	251	FRA	France	H5	Electrical machinery and equipment a	TRUE	\$3,896,779,636.89	\$-	\$3,896,779,636.89	4
22	20170401	2017	4	201704	392	JPN	Japan	H5	Electrical machinery and equipment a	TRUE	\$7,227,261,424.77	\$-	\$7,227,261,424.77	4
23	20170401	2017	4	201704	484	MEX	Mexico	H4	Electrical machinery and equipment a	FALSE	\$-	\$6,452,980,489.00	\$6,452,980,489.00	0
24	20170401	2017	4	201704	842	USA	USA	H5	Electrical machinery and equipment a	TRUE	\$27,644,507,252.00	\$27,237,686,836.00	\$27,644,507,252.00	4
25	20170501	2017	5	201705	124	CAN	Canada	H5	Electrical machinery and equipment a	FALSE	\$-	\$3,521,404,110.25	\$3,521,404,110.25	0
26	20170501	2017	5	201705	156	CHN	China	H5	Electrical machinery and equipment a	TRUE	\$34,598,625,296.00	\$-	\$34,598,625,296.00	4
27	20170501	2017	5	201705	251	FRA	France	H5	Electrical machinery and equipment a	TRUE	\$4,377,591,303.84	\$-	\$4,377,591,303.84	4
28	20170501	2017	5	201705	392	JPN	Japan	H5	Electrical machinery and equipment a	TRUE	\$7,326,696,068.22	\$-	\$7,326,696,068.22	4
29	20170501	2017	5	201705	484	MEX	Mexico	H4	Electrical machinery and equipment a	FALSE	\$-	\$7,655,981,786.00	\$7,655,981,786.00	0
30	20170501	2017	5	201705	842	USA	USA	H5	Electrical machinery and equipment a	TRUE	\$29,275,880,820.00	\$28,825,866,280.00	\$29,275,880,820.00	4
31	20170601	2017	6	201706	124	CAN	Canada	H5	Electrical machinery and equipment a	FALSE	\$-	\$3,569,224,546.75	\$3,569,224,546.75	0
32	20170601	2017	6	201706	156	CHN	China	H5	Electrical machinery and equipment a	TRUE	\$37,773,721,563.00	\$-	\$37,773,721,563.00	4
33	20170601	2017	6	201706	251	FRA	France	H5	Electrical machinery and equipment a	TRUE	\$4,650,915,831.41	\$-	\$4,650,915,831.41	4
34	20170601	2017	6	201706	392	JPN	Japan	H5	Electrical machinery and equipment a	TRUE	\$7,822,263,234.41	\$-	\$7,822,263,234.41	4
35	20170601	2017	6	201706	484	MEX	Mexico	H4	Electrical machinery and equipment a	FALSE	\$-	\$7,424,471,718.00	\$7,424,471,718.00	0
36	20170601	2017	6	201706										

Figure 3 : Data after Clean up and

The panel-style structure of the dataset allows for both time-series and cross-country analysis. Because the focus is on a specific commodity category, the dataset remains well-scoped and interpretable, ensuring that methodological insights are prioritized over data volume.

Key Variables

The primary dependent variable in this study is **Trade Value**, which represents the monetary value of imports for the selected commodity. These variables capture overall trade intensity and serve as the main indicator of disruption impact. Complementary variables such as CIF (Cost, Insurance, and Freight) and FOB (Free on Board) values provide additional valuation context and allow for robustness checks when interpreting trade movements.

Time-related variables play a critical role in the analysis. These include Reference Year, Reference Month, and a combined Period variable, which together enable trend modeling, seasonal adjustment, and structural break detection. These temporal identifiers are essential for constructing time-series regressions and cumulative change detection models.

Country-level identifiers and classification codes are also included to enable cross-sectional comparison. By combining country indicators with time variables, the dataset supports both within-country structural analysis and cross-country heterogeneity assessment.

Data Cleansing Process

Our effort required some initial data cleansing effort for our dataset. This was to drive toward an initial foundational dataset that we can begin to utilize for data exploration and modeling. Below are key steps taken to drive towards an initial cleanse/workable dataset :

1.0 - Data Loading & Integration

The raw trade dataset was imported from the UN Comtrade source file and consolidated into a structured analytical format. Country-level observations across multiple years were integrated into a unified panel dataset to enable consistent cross-country and time-series analysis.

2.0 - Data Standardization

Variable names, country labels, and time identifiers were standardized to ensure consistency across observations. Monetary values were formatted uniformly to maintain comparability and prevent scaling inconsistencies during modeling.

3.0 - Data Filtering

Irrelevant administrative fields, duplicate identifiers, and constant-value columns were removed to reduce dimensionality. The dataset was filtered to retain only the selected commodity category and target countries within the 2017–2023 timeframe.

4.0 - Preliminary Data Analysis

Initial summary statistics and time-series visualizations were generated to assess trends, seasonality, and volatility patterns. This step helped identify potential anomalies, synchronized declines, and early signs of structural instability.

5.0 - Data Type Mapping

Variables were mapped to appropriate data types, with numerical fields assigned as continuous variables and categorical fields encoded accordingly. Time variables were formatted to support chronological ordering and time-series modeling.

6.0 - Data Integration & Enrichment

Additional derived features, including time indices and seasonal indicators, were introduced to enhance modeling capability. The dataset was enriched to support anomaly detection, regression analysis, and structural break identification.

The data cleansing and preprocessing activities resulted in a refined, standardized, and enriched dataset. The cleansed dataset per the cleansing process described above resulted in 505 observations with 14 features. This will serve as a solid foundation for our analysis and act as our base data for modeling activities.

Exploratory Data Analysis & Findings

Before conducting anomaly detection and structural testing, we first examine aggregate trade behavior across all six countries to establish baseline patterns and cross-country differences. The following visualization presents monthly trade values over the full sample period.

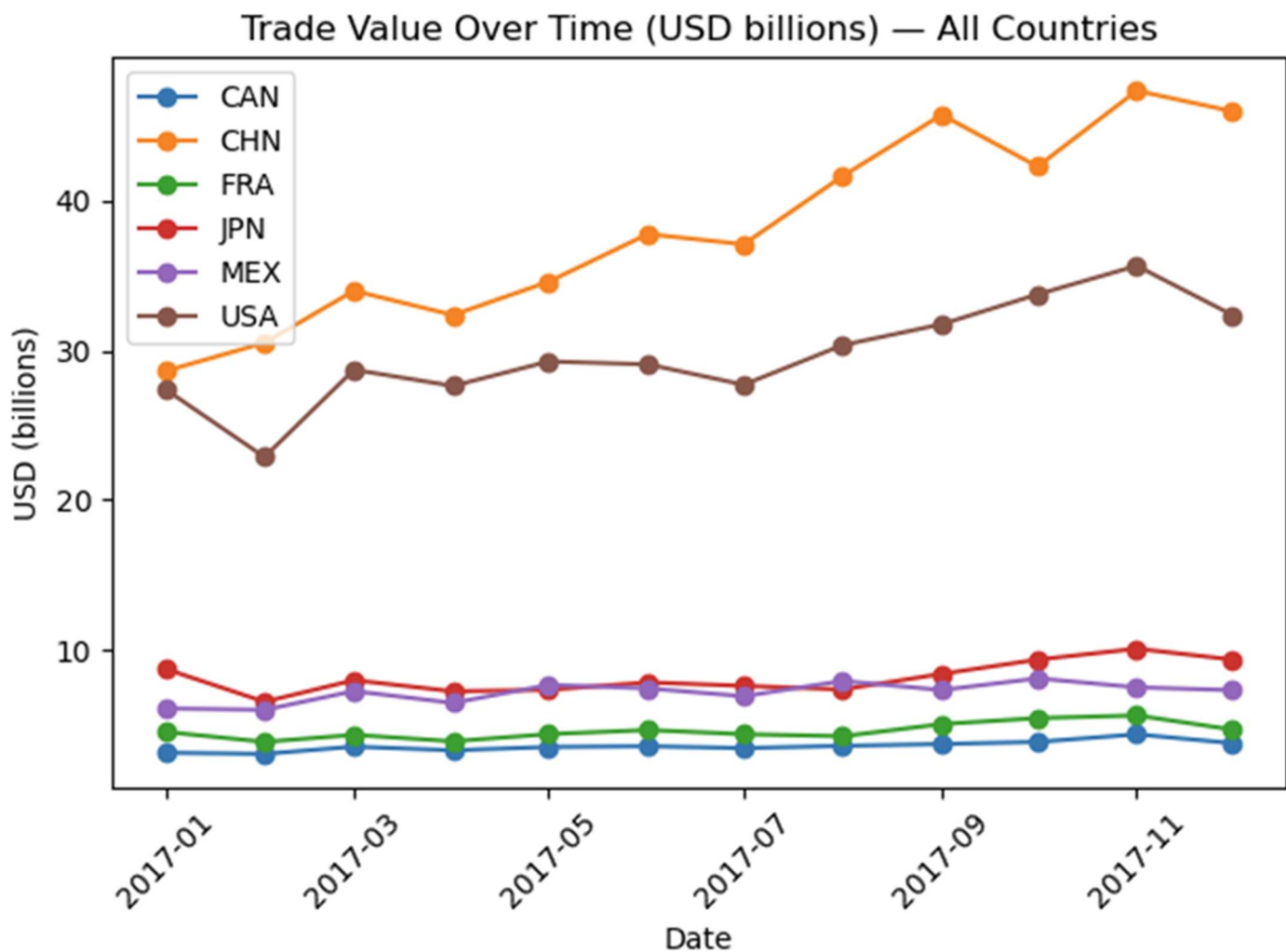
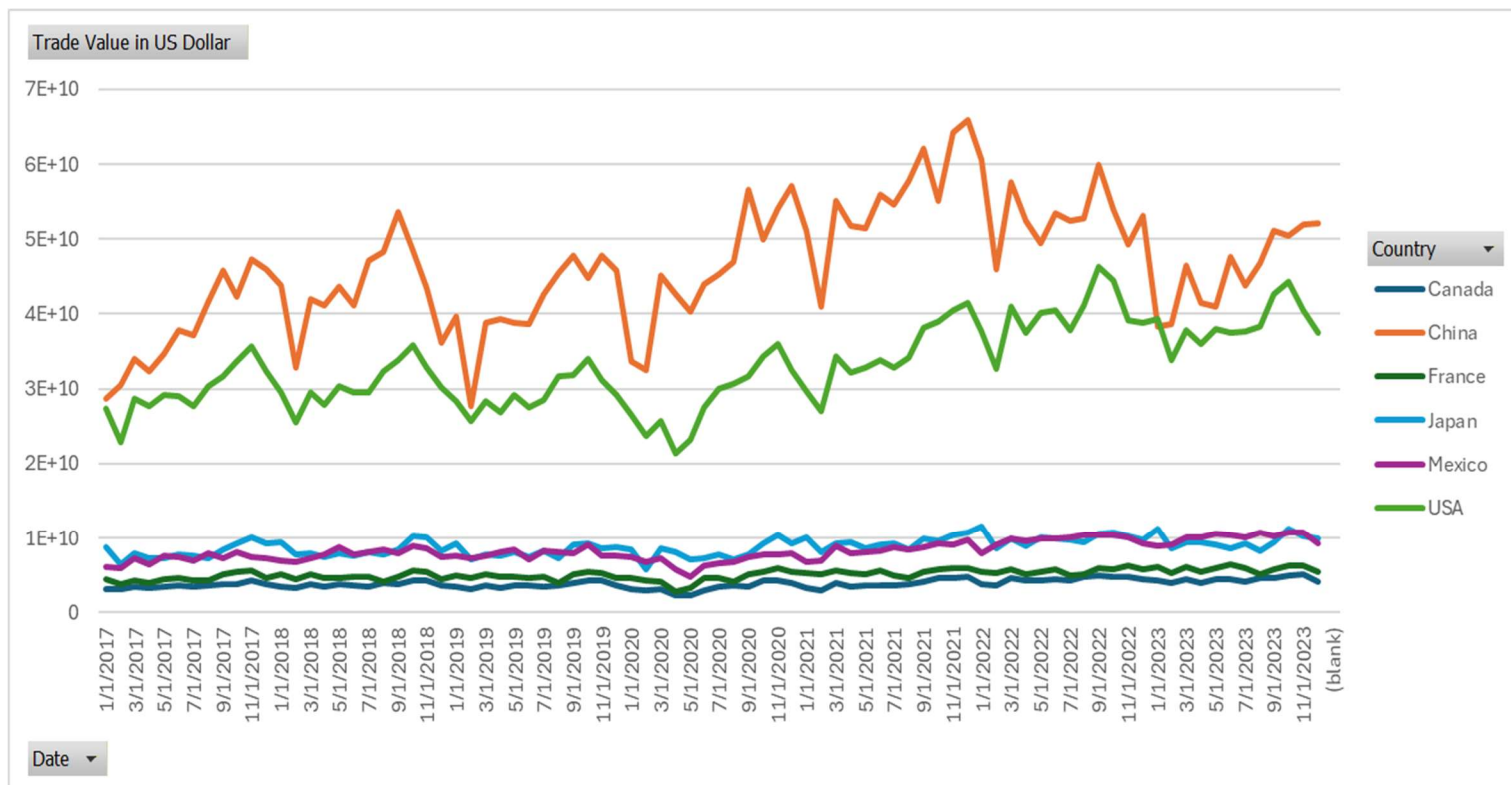


Figure 4 : Trade Value Over Time. all Countries.

The figure highlights substantial differences in trade scale across countries, with China and the United States exhibiting significantly higher trade volumes than Canada, France, Japan, and Mexico. Seasonal patterns are visible across most economies, reinforcing the importance of controlling trends and seasonality in baseline modeling. A pronounced contraction appears in early 2020 across all countries, suggesting a synchronized global disruption event. These preliminary observations motivate the use of standardized residual-based methods to ensure meaningful cross-country comparison independent of trade size.



Initial time-series visualization reveals a clear long-term upward trend in trade values for most countries prior to 2020. China consistently records the highest trade levels, reflecting its dominant role in global manufacturing and supply chain integration. Across countries, seasonal fluctuations are visible, indicating recurring monthly patterns in trade activity.

A synchronized and pronounced contraction is observed in early 2020 across all six countries. This clustering of negative deviations aligns with COVID-19–related lockdowns, production stoppages, and global logistics disruptions. The magnitude and simultaneity of this decline suggest a structural shock rather than normal volatility.

Following 2020, a recovery phase emerges during 2021–2022, though the speed and volatility of recovery differ by country. Some economies exhibit smoother rebounds, while others show persistent volatility spikes. These findings provide preliminary evidence of structural instability and motivate the application of formal anomaly detection and structural change models.

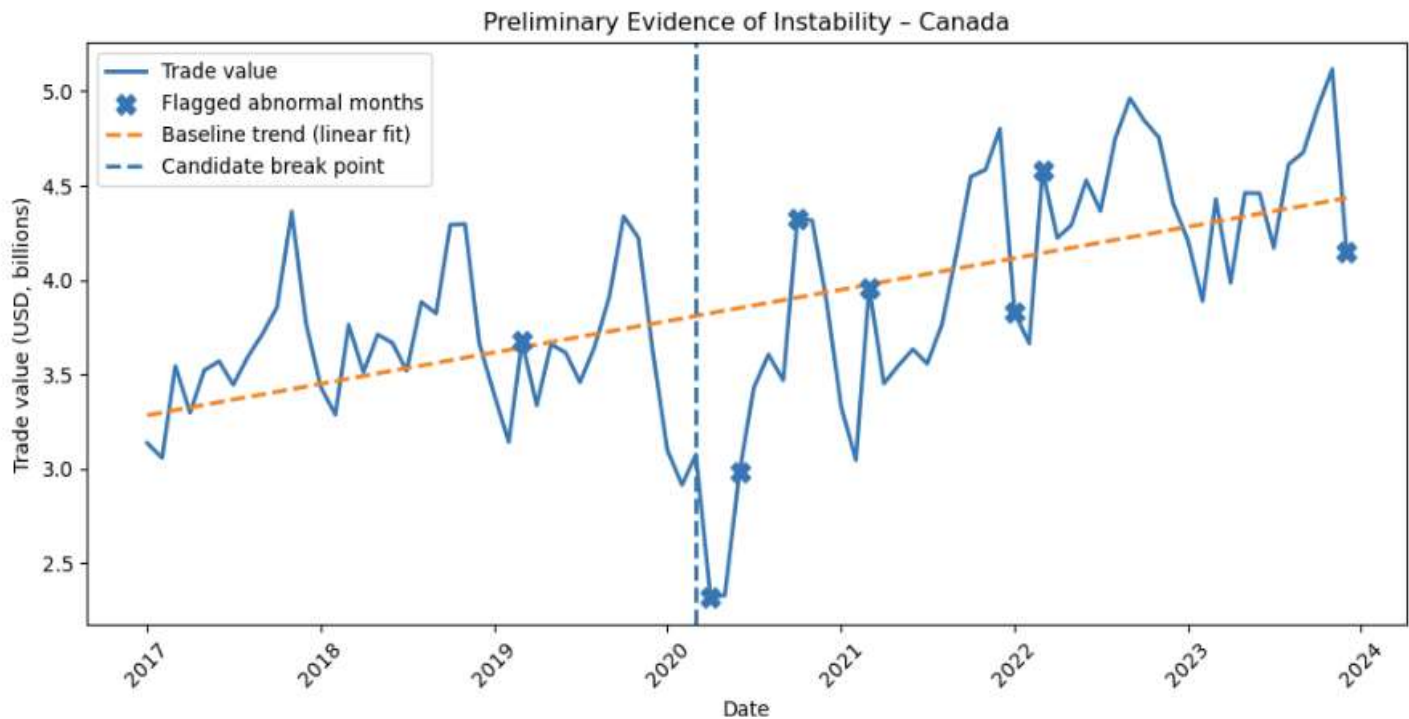


Figure 6 : Preliminary Evidence of Instability in Canada

Feature Engineering

To enhance model performance and interpretability, several engineered features were introduced. A time index variable was created to represent the linear progression of months, enabling trend estimation in baseline regression models. Seasonal indicators were also constructed to capture recurring monthly effects.

For anomaly detection, standardized trade values (Z-scores) were computed to quantify deviations from historical norms. This transformation allows extreme months to be identified using consistent statistical thresholds across countries. Residual-based transformations were later used in CUSUM analysis to detect persistent cumulative shifts.

Additionally, regime indicators were introduced in piecewise regression models to differentiate pre-disruption, disruption, and post-disruption phases. These engineered features enable the model to explicitly capture structural breaks and compare trade dynamics across economic regimes.

Challenges Encountered

One of the primary challenges encountered in this study was balancing data completeness with analytical clarity. The raw dataset contained numerous administrative and metadata fields that were not directly relevant to modeling trade variability. Distinguishing between economically meaningful variables and system-generated

reporting indicators required careful review to avoid either discarding useful information or introducing unnecessary noise into the analysis. Additionally, ensuring consistent time alignment across countries was essential to prevent biased comparisons in structural break detection.

Another challenge involved differentiating true structural instability from normal seasonal fluctuations. Because trade data inherently exhibits recurring monthly patterns, abrupt deviations during disruption periods could potentially be misinterpreted without properly accounting for trend and seasonality. This required a disciplined approach to baseline modeling before applying anomaly detection and change-point methods. Finally, cross-country heterogeneity posed interpretive complexity, as differences in volatility and recovery patterns needed to be analyzed within both a statistical and economic context rather than relying solely on quantitative thresholds.

OVERVIEW OF MODELS

1. Models Used & Cross-Platform Comparison

1.1 Modeling Framework Overview

To ensure methodological robustness, transparency, and reproducibility, the modeling framework combines several complementary techniques. The core structural models include baseline time-series regression, residual-based anomaly detection (Z-score), piecewise regression, and CUSUM change detection. These models are used to capture the timing, magnitude, and persistence of trade disruptions.

In addition, a lag-based model was implemented to improve predictive accuracy by capturing short-term autocorrelation in trade flows. While this model enhances forecasting performance, it is not part of the core structural detection framework.

This layered approach allows structural instability to be evaluated from multiple statistical perspectives, reducing the risk that detected disruption signals arise from assumptions specific to any single model.

Summary of Models Used in the Structural Detection Framework

Model	Type	Purpose
Baseline Regression	Structural Analysis	Establish trend and seasonality

Z-Score Detection	Structural Analysis	Identify extreme deviations
Piecewise Regression	Structural Analysis	Detect structural breaks
CUSUM	Structural Analysis	Measure the persistence of instability
Lag Model	Predictive Support	Improve predictive accuracy (autocorrelation)
ARIMA	Predictive Time Series Modeling	Capture autocorrelation and temporal dynamics to evaluate deviations

1.2. Sensitivity Testing

Sensitivity testing was conducted by varying the anomaly detection threshold from a strict level ($|Z| \geq 2.5$) to a more moderate level ($|Z| \geq 2.0$). This allows evaluation of how detection results change under different statistical cutoffs and helps assess the robustness of identified structural deviations.

Under the moderate threshold, additional deviations are detected across both disruption and non-disruption periods, indicating higher sensitivity but lower structural specificity. In contrast, the strict threshold isolates extreme, shock-driven events concentrated in early 2020.

For example, in the case of the United States, the April 2020 deviation falls just below the strict threshold ($Z \approx -2.49$) and is therefore not classified as an anomaly under the stricter specification. However, under the moderate threshold ($|Z| \geq 2.0$), this period is flagged as abnormal, illustrating how threshold selection affects detection sensitivity while maintaining the broader interpretation of disruption timing.

1.3. Cross-Platform Validation (Python vs R)

The modeling framework was implemented in two independent analytical environments: Python and R. Python served as the primary platform for constructing the structural validation pipeline, while R was used for independent validation using alternative statistical libraries.

The use of two analytical environments strengthens internal validity by reducing the risk of software-specific bias, implementation error, or package-dependent assumptions. Outputs from both implementations were compared to verify consistency in the timing and direction of detected disruptions.

Where both implementations converge, confidence increases that the detected instability reflects economic behavior rather than computational artifacts. This cross-platform validation enhances reproducibility and aligns with best practices in empirical economic research.

2. Model Selection & Threshold / Validation Design

2.1. Model Selection Philosophy

Model selection in this study intentionally departs from traditional forecasting-centric evaluation. While predictive accuracy metrics such as RMSE and MAE were computed as reference diagnostics, they were not the primary criteria for model validation. In disruption contexts, low prediction error does not necessarily imply structural adequacy, as a model may fit historical data well while failing to detect meaningful regime shifts.

Instead, the focus is placed on identifying structural instability, including the timing, magnitude, and persistence of disruptions.

2.2. Structural Validation Criteria

Rather than relying solely on forecasting accuracy, models were evaluated using a structural validation framework designed for disruption settings:

- **Timing Consistency Across Countries:** Do detected deviations align with known global disruption periods?
- **Persistence of Deviations:** Do shocks accumulate into structural shifts, or do they revert quickly?
- **Cross-Country Structural Alignment:** Are disruption signals synchronized or staggered across economies?
- **Noise Control:** Does the model avoid flagging routine volatility as structural instability?

This framework prioritizes interpretability and structural relevance over pure forecasting performance.

2.3. Literature Support

This approach is grounded in the structural break literature. Zeileis et al. (2002) argue that:

“Economic time series often experience structural change during crises, and ignoring such breaks leads to misleading inference.”

Similarly, Perron¹¹ demonstrates that ignoring structural breaks can lead to biased inference and incorrect conclusions about persistence and trend stability. These findings reinforce the importance of explicitly testing for regime changes in disruption-driven environments.

2.4. Threshold Design & Robustness

By incorporating anomaly detection, piecewise regression, and cumulative residual analysis, the modeling strategy accounts for regime-dependent behavior. Key validation choices, such as the Z-score threshold, were selected based on economic interpretability and robustness rather than arbitrary statistical cutoffs.

A strict threshold ($|Z| \geq 2.5$) was used to isolate extreme deviations associated with systemic shocks, while a moderate threshold ($|Z| \geq 2.0$) was used for sensitivity testing. This allows evaluation of how detection changes under different levels of statistical strictness and helps distinguish true structural breaks from routine volatility.

The final model selection reflects a balance between statistical rigor and economic meaning. Rather than optimizing for minimal forecast error, the framework prioritizes structural insight, specifically the detection, timing, and persistence of disruption-induced instability.

3. R Models Section

To evaluate structural instability in international trade flows, a sequence of complementary econometric models was implemented in the R statistical environment. Rather than relying on a single modeling technique, the analysis applies a layered approach that combines baseline trend estimation, anomaly detection, structural break modeling (Piecewise regression), a lag model to capture how previous trade values influence current trade levels, cumulative residual diagnostics (CUSUM) and ARIMA based time series model.

These models are consistent with the overall modeling framework described earlier, where baseline regression, Z-score detection, piecewise regression, and CUSUM are used for structural analysis, while the lag model is included as a supporting model to improve predictive accuracy. This multi-model strategy allows the analysis to capture different dimensions of disruption behavior, including extreme short-term deviations, regime shifts in trend dynamics, and the persistence of structural disturbances over time.

$$Trade_t = \beta_0 + \beta_1 Time_t + \sum \gamma_m Month_{m,t} + \epsilon_t$$

Parameters: β_0 (level), β_1 (trend), γ_m (seasonality), σ^2 (variance)

Where **$Trade_t$** represents the trade value at time **t** , **t** is the time trend, $Month_m$ captures seasonal month effects, and **ϵ_t** represents the error term.

To detect extreme short-term outliers, Z-score anomaly detection was implemented, which standardized the trading values relating to their means and standard deviations. Observations that exceeded ± 2.5 standard deviations are considered anomalies. This approach identifies sudden spikes or drops in trade flows.

$$Z_t = \frac{(Trade_t - \mu)}{\sigma}$$

$|Z_t| > 2.5$

Parameters: μ and σ

Piecewise regression: is utilized as it incorporates structural breakpoints indeed through testing. When a breakpoint is detected, the model changes in slope and levels after the breakpoint time. This framework models regime transition rather than considering them as random noise. As a result, piecewise regression provides a clearer structural interpretation of disruption effects and improves overall model fit.

$$Trade_t = \beta_0 + \beta_1 t + \beta_2 PostTime_t + \sum \gamma_m Month_{m,t} + \epsilon_t$$

Parameters: β_0 (level), β_1 (trend), β_2 (change), γ_m (seasonality), σ^2 (variance)

A Lag Model: was utilized to improve predictive accuracy, which goes beyond the Piecewise and time trend model, which incorporates autocorrelation into the trade series. Trade flows often show persistence in the short term, and the lagged trade value model will capture autocorrelation that a time trend cannot fully account for. This will provide improved predictive accuracy and simplicity.

$$Trade_t = \beta_0 + \beta_1 \times Tade + t_{-1} + \sum_{12}^{m=1} \gamma_m \times Month_m + \epsilon_t$$

Parameters: β_0 (level), β_1 (lag), γ_m (seasonality), σ^2 (variance)

To evaluate instability and persistence, CUSUM was applied. CUSUM accumulates regression residuals over time and evaluates whether the cumulative path crosses predefined boundaries. Sustained upward or downward movements signal structural instability. This approach allows differentiation between isolated volatility and persistent deviations.

$$CUSUM_t = \sum_{i=1}^t Residual_i$$

Parameters: Residuals from the parametric baseline model and Threshold

ARIMA-Based Time Series Model: was utilized is designed specifically for time series data, unlike regression-based models. ARIMA captures random fluctuation and temporal dependence by utilizing past values and errors to current trade levels. ARIMA is suitable for real world trade series that exhibit trends and behavior that evolve over time. ARIMA is a robust approach which adaptive for predicting trade levels, leveraging random fluctuations and historical data

$$y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \Theta_j \epsilon_{t-j} + \epsilon_t$$

Parameters: c (level), p (autoregressive order), d (differecing), q (moving average order), ϕ_i (auto regressive effects), ϑ_j (moving average effects), σ^2 (variance)

Together, these modeling components form a layered structural detection framework. The baseline regression establishes the expected trade trajectory under normal conditions, while Z-score anomaly detection isolates extreme short-term deviations. Piecewise regression then models regime transitions by allowing trend adjustments after structural breakpoints, and CUSUM analysis evaluates whether deviations accumulate into persistent instability over time.

By combining point-based anomaly detection with cumulative persistence diagnostics, the framework provides a more comprehensive assessment of disruption dynamics than any single method alone.

Baseline trend regressions with seasonal controls are commonly used in economic time-series analysis to establish a reference trajectory against which abnormal deviations can be measured. As noted by Hamilton⁸, modeling underlying trends and cyclical components provides a necessary benchmark for identifying nonstationary behavior and structural instability in macroeconomic time series.

4. R Model Evaluation & Structural Results

4.1 Z-Score Anomaly Detection :

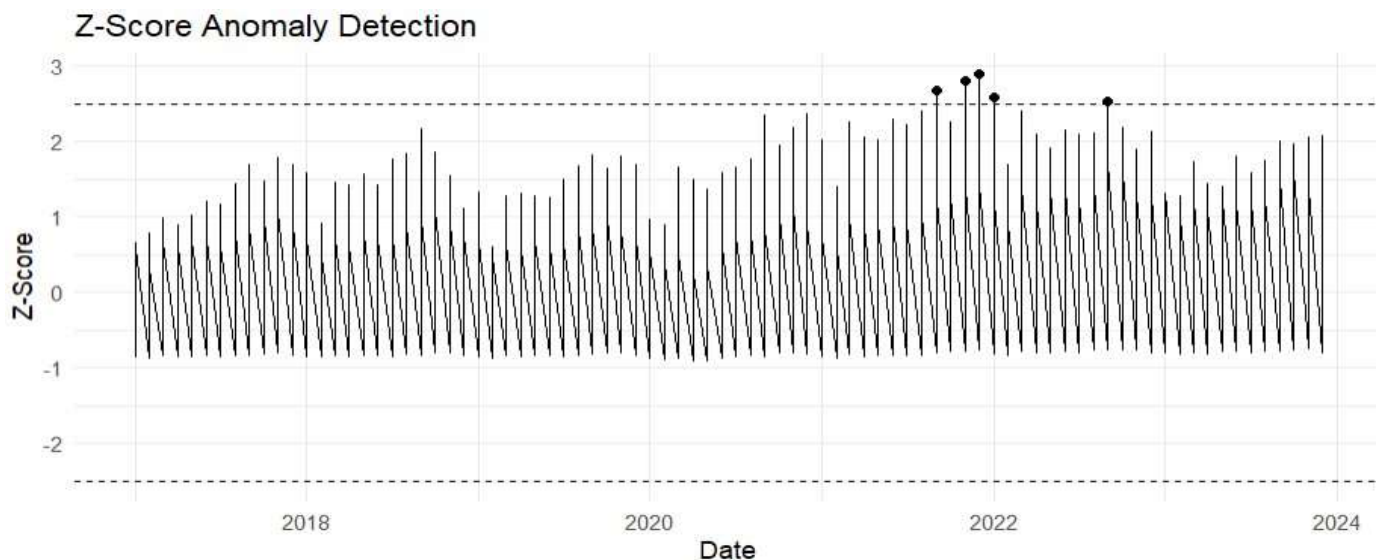


Figure 7: Z-Score Anomaly Detection of Trade Values (2017–2023)

The graph reveals clustering of extreme negative and positive values beginning in 2020. Rather than appearing as isolated spikes, anomalies occur in concentrated periods, indicating sustained abnormal activity. This pattern suggests that the pandemic shock was not a single-month disturbance, but a period of heightened volatility.

Z-score detection highlights the magnitude of short-term shocks; however, it does not determine whether the underlying trade-generating process has permanently shifted. In later years, the frequency of extreme Z-score values declines, suggesting a gradual stabilization in trade patterns.

4.2 Trade Over Time with Piecewise Trend

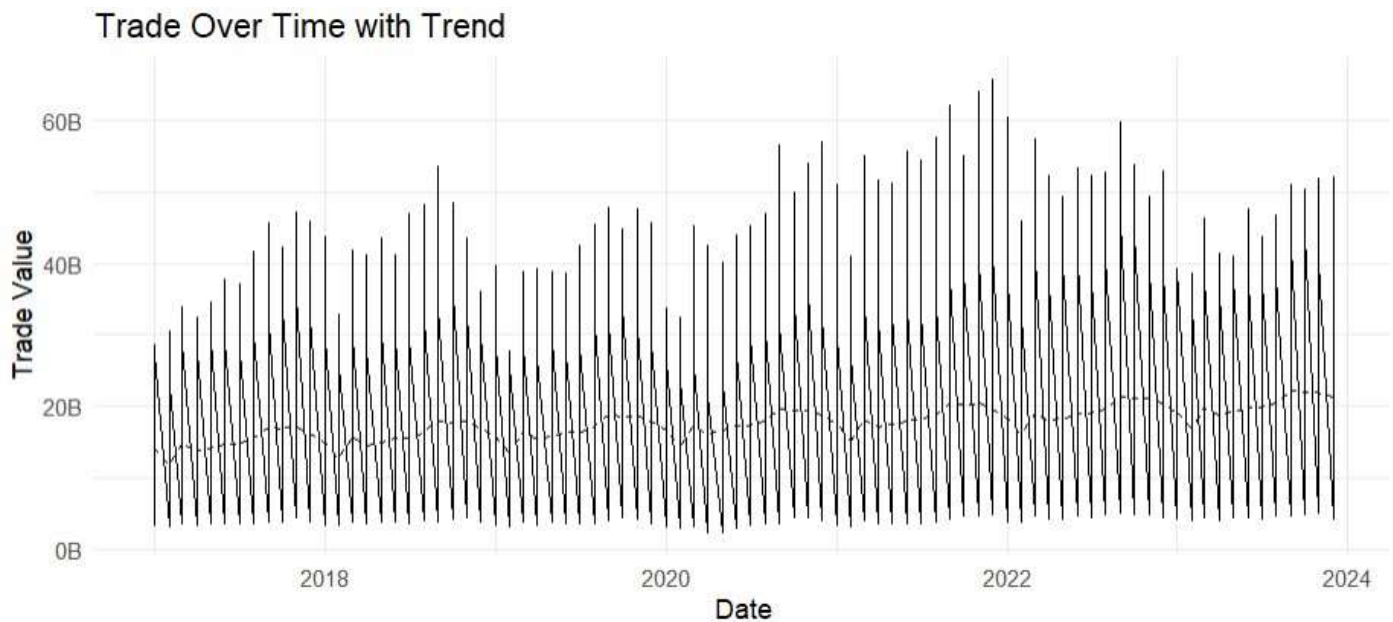


Figure 8: Trade Values with Piecewise Regression Trend and Structural Breaks

The trade over time plot shows the import values along with the fitted piecewise regression trend. The graph displays a long-term upward trajectory prior to 2020, accounting for seasonal fluctuations across years. Around 2020, there is a visible structural disturbance. Trade values decline sharply relative to previous years. This break aligns with the problems of the COVID-19 pandemic, which introduced supply chain disruption, transportation issues, and more. The fitted piecewise regression captured the shifts, allowing the slope to shift after detecting a breakpoint. After the disruption period, trade recovers, but the slope characteristics differ from before the 2020 era, suggesting trade did not ever return to its original path but transitioned into a different phase of development.

Structural break models are particularly valuable in crisis-driven environments where economic relationships may shift abruptly. Bai and Perron² show that allowing for multiple structural breakpoints significantly improves the ability of regression models to capture regime changes in economic time series.

4.3 CUSUM Change Detection Analysis

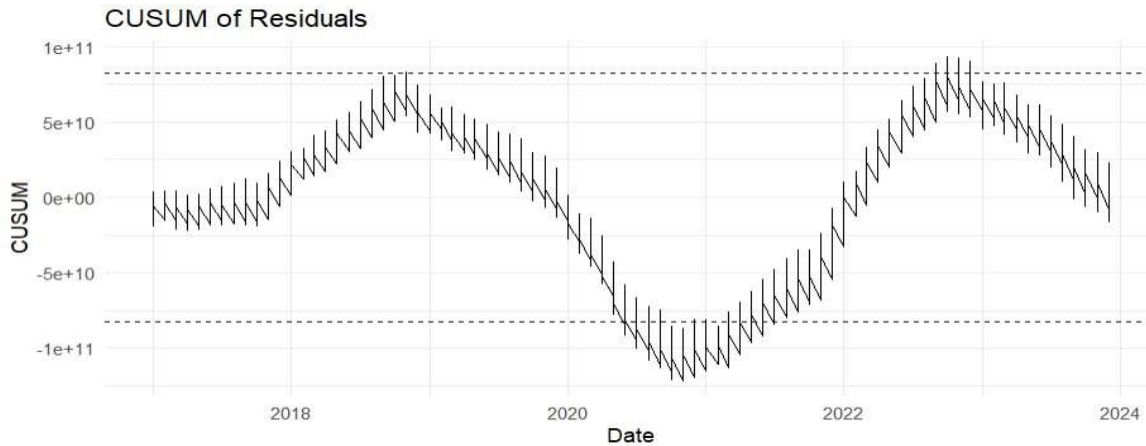
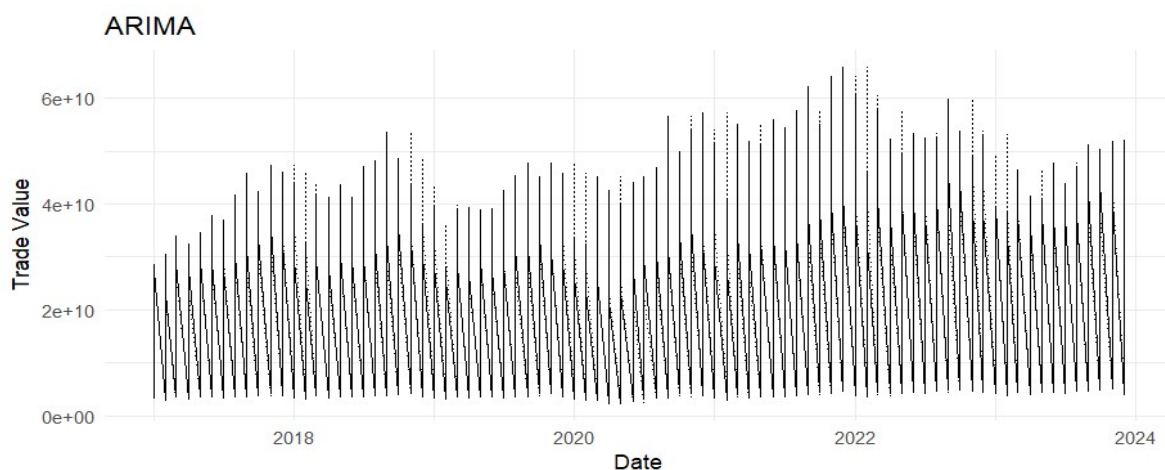


Figure 9: CUSUM of Residuals for Structural Change Detection

As shown in the Figure above, the CUSUM plot tracks the accumulated deviations of observed trade values from the piecewise regression model. The plot shows a downward movement beginning in 2020, which crosses thresholds and signals extreme deviations from the established trends. This aligns with the start of the COVID-19 pandemic. The negative accumulation reflects the consecutive negative residuals, suggesting trade values fell below model expectations during the disruption period. A strong upward correction occurs around 2022 to 2023, which suggests positive residuals.

The CUSUM plots detected persistent model deviations caused by structural change. Residual-based cumulative diagnostics such as CUSUM are widely used to detect parameter instability and persistent structural change. Brown, Durbin, and Evans⁵ demonstrate that cumulative residual analysis provides a powerful method for identifying sustained deviations from model expectations, particularly during periods of economic disruption or policy shocks.

4.4 ARIMA-Based Time Series Model Plot



The ARIMA plot shows observed trade values along with the fitted values. The line follows the trade series, which captures upward trends and fluctuations. From the graph, it is seen that the model adapts well to changes in trade. The model peaks smoothly, which shows it captures the underlying time dependence. There are strong spikes in points in time, which can be due to seasonal demand. There is noticeable disruption from 2020 to 2022 with irregular peaks and increased volatility which is due to the effects of the pandemic, which caused global disruption. Trade values increased from 2021 to 2022 and reached their highest level. After 2022, values stabilize. The ARIMA fitted values track movement effectively.

5. Model Evaluation

5.1 Model Evaluation Summary

Model Metric	Result	Model	Description
Z-Score Flags	5	Z-Score Detection	Number of observations exceeding ± 2.5
CUSUM Signals	63	CUSUM	Periods where residuals crossed stable thresholds
RMSE (%)	94.02	Piecewise Regression	Root Mean Squared Error
MAE (%)	83.76	Piecewise Regression	Mean Absolute Error
RMSE (%)	78.78	Lag Model	Root Mean Squared Error
MAE (%)	67.03	Lag Model	Mean Absolute Error
RMSE (%)	18.4	ARIMA	Root Mean Squared Error
MAE (%)	10.1	ARIMA	Mean Absolute Error

The performance of anomaly detection and structural models was evaluated using statistical approaches and error metrics. The results provide insightful observations into trends and the overall performance of the models. Model performance for piecewise regression, lag-based model, and ARIMA was evaluated using RMSE and MAE in both absolute terms and as a percentage of mean trade values. The RMSE and MAE values reported in the model evaluation summary are percentages of the mean, while anomaly detection metrics are reported as counts.

The Z-score direction identified 5 observations that exceed the ± 2.5 standard deviation. These outliers exhibit extreme deviations from the average values and are considered short-term anomalies in trade. The small number of deviations suggests outliers are rarer than frequent, and when visualized, it can be seen concentrated around 2021 -2022, which corresponds to the disruption caused by the COVID-19 pandemic.

The CUSUM evaluation produced 63 signals where the residuals crossed boundaries, indicating continuous and persistent deviations. CUSUM is designed to detect a shift in the data generating process. This high number of signals suggests periods of disruption in the trade. The CUSUM plot highlights the deviation state around 2020, which reflects on the instability of trade during the pandemic. The upward trend after the downward suggests a recovery phase where trade values began to exceed what the model expects.

The model accuracy for the piecewise regression model was evaluated using Root Mean Squared Error and Mean Absolute Error, where both have high error, with RMSE being 94 and MAE being 83.8. This reflects the month to month volatility that a time-based model cannot fully capture.

The model accuracy of the Lag model was evaluated, and the results show RMSE being 78.8, which suggests that incorporating previous month trade values reduces RMSE, showing that past trade values influence current trade levels. MAE was 67, which shows absolute errors decrease substantially as well, which shows the lag model captures fluctuations better. The Lag model improves predictions by leveraging autocorrelation.

The ARIMA model performs significantly better than other models utilized, with much lower RMSE and MAE values with RMSE being 18.4 and MAE being 10.1. It is best performing because it captures time dependencies of the data by incorporating past values and errors into predictions. ARIMA could also handle autocorrelation and non-stationary behaviours effectively. ARIMA provides a more accurate and reliable fit for trade data, and also fully represents the time series structure.

6. Python Model Results

To illustrate how the models detect structural disruptions, we now present the results of the anomaly detection analysis under different threshold settings.

6.1. Strict Anomaly Threshold ($|Z| \geq 2.5$)

Under the strict Z-score threshold ($|Z| \geq 2.5$), abnormal trade deviations are highly concentrated in early 2020.

Abnormal Trade Deviations Under Strict Z-Score Threshold ($ Z \geq 2.5$)							
	country	model_name	series	break_date	value	zscore	months_from_event
0	CHN	ResidualZ	primaryValue	2020-02-01	3.254526e+10	-2.537111	-1
1	JPN	ResidualZ	primaryValue	2020-02-01	5.831429e+09	-2.925708	-1
2	CAN	ResidualZ	primaryValue	2020-04-01	2.316924e+09	-2.523024	1
3	FRA	ResidualZ	primaryValue	2020-04-01	2.809751e+09	-3.737705	1
4	FRA	ResidualZ	primaryValue	2020-05-01	3.246765e+09	-2.800882	2
5	MEX	ResidualZ	primaryValue	2020-05-01	4.843716e+09	-3.517732	2

The detected abnormal events across countries are summarized as follows:

- China: 1 abnormal month (February 2020)

- Japan: 1 abnormal month (February 2020)
- Canada: 1 abnormal month (April 2020)
- France: 2 abnormal months (April–May 2020)
- Mexico: 1 abnormal month (May 2020)

United States: 0 abnormal months (April 2020 $Z = -2.49$, just below the threshold)

6.2. Propagation of Abnormal Trade Events ($|Z| \geq 2.5$)

The below plot visualizes the timing of abnormal trade deviations across countries, showing that disruptions occurred in a staggered sequence rather than simultaneously.

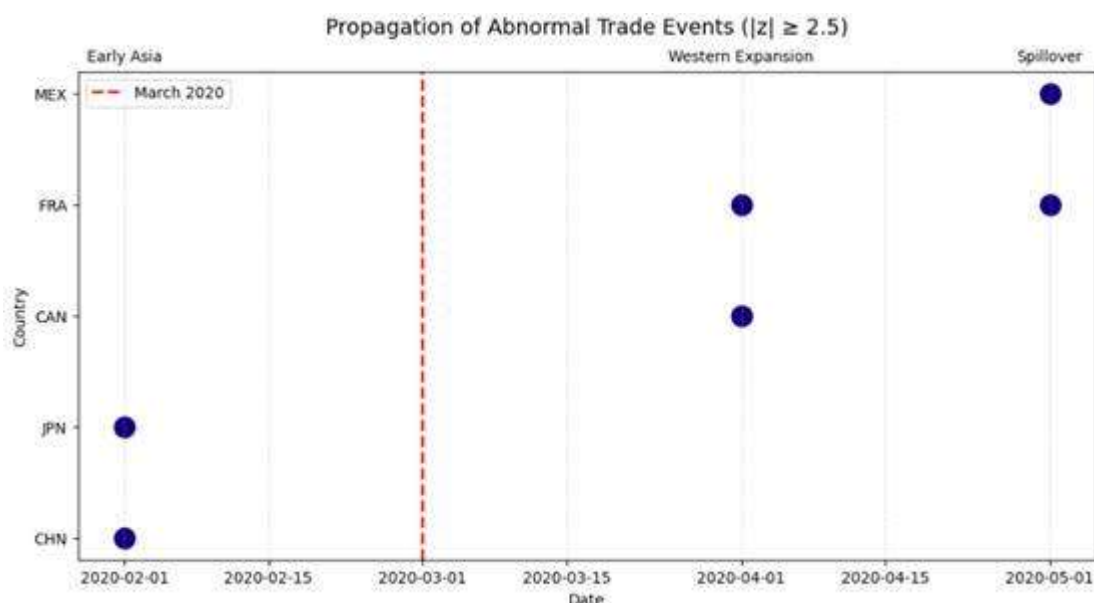


Figure 10: Propagation of Abnormal Trade Events Across Countries ($|Z| \geq 2.5$)

Abnormal deviations between February and May 2020, indicating staggered structural transmission.

6.3. Structural Interpretation

The propagation timeline reveals a temporally concentrated yet staggered sequence of structural deviations across countries. Rather than observing simultaneous collapse across all six economies, the pattern suggests 3 phase transmissions :

Phase 1: Early Regional Impact (February 2020)

China and Japan exhibit the earliest extreme deviations. This timing is economically intuitive. The COVID-19 outbreak originated in East Asia, leading to early factory shutdowns, port disruptions, and supply chain interruptions.

The February deviations in China and Japan are consistent with a regionally concentrated first-stage shock.

This observation aligns with Baldwin and Freeman³, who argue:

“Supply chains propagate shocks in waves, creating contagion patterns across countries rather than simultaneous collapse.”

This wave-like transmission mechanism is also consistent with network-based macroeconomic theory. Acemoglu et al.¹ demonstrate that production networks can amplify and transmit shocks through inter-industry linkages, generating non-linear and staggered adjustment across economies. In highly interconnected trade systems, disturbances in one region can propagate indirectly through supplier and demand relationships rather than appearing as simultaneous collapses.

The staggered deviations observed between February and May 2020 align with this network amplification mechanism, suggesting that global trade instability reflected interconnected supply chain exposure rather than isolated national shocks.

Phase 2: Western Transmission (April 2020)

Canada and France show structural deviations approximately one month later. This timing corresponds with widespread lockdown measures implemented in Europe and North America during March and April 2020.

Bonadio et al.⁴ note:

“Global supply chain linkages amplified the economic impact of COVID-19 across countries.”

The April breakpoints suggest that trade disruption is transmitted through global production networks, consistent with supply-chain amplification mechanisms documented in the literature.

Phase 3: Secondary Spillover (May 2020)

France (again) and Mexico exhibit abnormal deviations in May 2020, suggesting that adjustment dynamics continued beyond the initial transmission wave. This may reflect delayed industrial shutdowns, lagged demand contraction, or inventory correction effects.

The IMF⁹ reports that:

“Trade volumes contracted sharply in early 2020, with uneven recovery across regions.”

This recovery pattern is further supported by research on global value chain resilience theory. Miroudot¹⁰ argues that while global supply networks are vulnerable to shock transmission, they also exhibit adaptive adjustment mechanisms that allow reconfiguration and recovery rather than permanent fragmentation.

The post-2021 stabilization observed in the CUSUM trajectories suggests that the COVID-19 shock, though severe, did not fundamentally dismantle trade regimes. Instead, it triggered temporary structural disruption followed by gradual system rebalancing.

The May observations support this uneven recovery narrative.

Collectively, the three-phase structure suggests that global trade instability during early 2020 followed a network diffusion pattern rather than a synchronized collapse. Initial region-specific disruption in East Asia

propagated to Western economies with a measurable lag, and subsequent spillovers reflected adjustment dynamics rather than immediate system-wide breakdown.

This phased structure strengthens the interpretation that structural instability was transmitted through global production linkages rather than emerging independently across countries.

Concentration and Structural Significance

Importantly, all strict-threshold deviations occur within a three-month window (February–May 2020). No additional extreme deviations are detected outside this interval.

This concentration strongly supports the interpretation of:

- A systemic global disruption
- A temporally bounded structural break
- A shock-driven instability rather than persistent volatility

If structural instability were ongoing or regime-shifting in nature, abnormal months would be distributed more widely across the time horizon. Instead, the clustering indicates a shock-centered regime disturbance.

This finding is consistent with the structural break literature (Zeileis et al.¹³), which emphasizes that crisis periods generate identifiable, concentrated breakpoints rather than diffuse instability.

The Case of the United States

Under the strict threshold, the United States does not register an abnormal month. However, the April 2020 Z-score (-2.49) falls marginally below the cutoff.

This nuance is analytically important.

The absence of a strict anomaly does not imply that the U.S. was unaffected. Rather, it suggests that:

- The deviation, while significant, was not extreme relative to historical volatility.
- Larger and more diversified economies may absorb shocks differently.
- Aggregate trade data may mask sector-specific collapses.

This interpretation is reinforced by the CUSUM persistence analysis, where the U.S. exhibits cumulative structural drift despite lacking a single extreme monthly deviation. Thus, anomaly detection and cumulative deviation analysis together provide a more complete structural narrative.

6.4. Economic Implications

The strict anomaly results indicate that:

1. Structural trade instability was concentrated and temporally bounded.
2. Shock transmission occurred in staggered waves rather than instantaneously.
3. Recovery dynamics varied across economies.

4. Extreme deviations were short-lived rather than persistent.

These findings suggest that the COVID-19 shock, while severe, did not permanently alter aggregate trade regimes for the selected commodity. Instead, trade patterns reverted toward baseline trajectories following the initial disruption phase.

6.5. Propagation and Magnitude

The severity-colored visualization adds an important intensity dimension to the propagation narrative. While all flagged months exceed the strict structural threshold ($|Z| \geq 2.5$), the magnitude of deviations varies across countries.

France's April 2020 deviation represents one of the largest absolute residual shocks in the sample, indicating a particularly severe disruption during the Western transmission phase. Mexico's May deviation is similarly pronounced, suggesting a strong secondary spillover effect. China and Japan exhibit substantial early shocks in February 2020, consistent with the regional origin of the disruption. In contrast, Canada's deviation, while statistically significant, lies closer to the threshold, indicating a comparatively moderate structural impact.

The color gradient therefore reflects the relative extremity of deviations: darker values indicate larger departures from historical baseline behavior, whereas lighter values remain closer to the anomaly cutoff. This distinction reinforces that the disruption narrative is not only temporal but also intensity based.

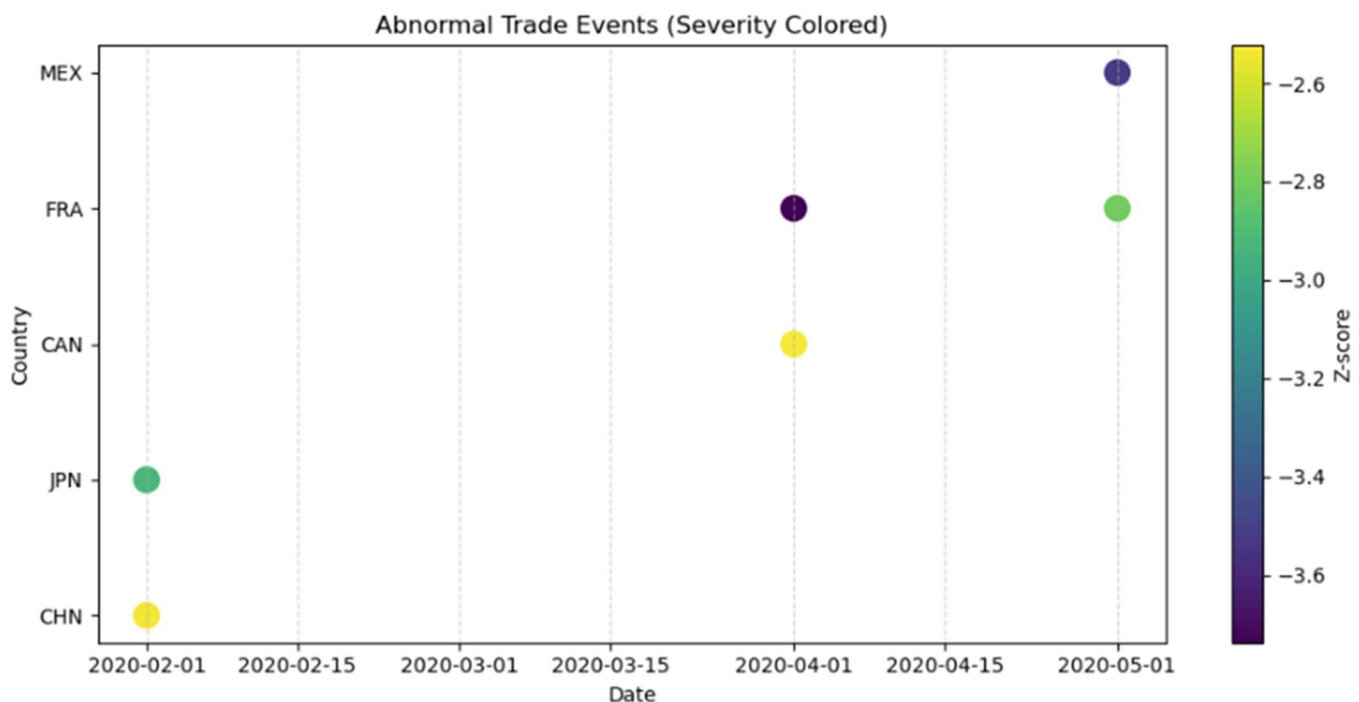


Figure 11: Severity of Abnormal Trade Events by Country (Z-Score Magnitude)

The timing pattern reveals three distinct disruption phases, consistent with wave-like transmission dynamics rather than simultaneous collapse.

Concentration and Structural Significance

The severity-colored visualization complements the earlier timing analysis by illustrating differences in the magnitude of structural deviations across countries. While the previous section established the phased propagation of disruption events, the severity plot highlights how intensely each country experienced the shock. France and Mexico exhibit some of the largest deviations relative to historical trade patterns, suggesting deeper disruption during the Western transmission and spillover phases. China and Japan show strong early deviations consistent with the regional origin of the shock.

Canada's deviation, although statistically significant, remains closer to the anomaly threshold, indicating a comparatively moderate structural impact.

Large absolute Z-scores indicate unusually large departures from historical trade patterns. Countries with stronger deviations may therefore have experienced deeper supply-chain exposure or weaker shock-absorption capacity. Thus, the propagation narrative is not only temporal but also intensity-based.

6.6. Sensitivity Analysis ($|Z| \geq 2.0$)

To evaluate robustness, the anomaly threshold was lowered from $|Z| \geq 2.5$ to $|Z| \geq 2.0$. Under the moderate threshold, abnormal months are detected not only in 2020 but also throughout 2017–2022. As seen below, China jumped from 1 → 9 events under the lower threshold. The disproportionate increase in flagged events for China under the moderate threshold suggests higher baseline volatility relative to other countries.

This indicates that lower thresholds may over-identify deviations in economies with inherently higher variance.

Strict vs Moderate Event Counts

Country	Events ($z \geq 2.5$)	Events ($z \geq 2.0$)
CAN	1	4
CHN	1	9
FRA	2	4
JPN	1	4
MEX	1	4
USA	0	5

Under the moderate threshold:

- The number of flagged months increases substantially across all countries.
- Events are distributed across both disruption and non-disruption periods.

Interpretation of Sensitivity Results: Lowering the anomaly threshold increases detection sensitivity but reduces structural specificity, leading to a higher likelihood of capturing routine volatility alongside true regime shifts.

In practical terms:

- The strict threshold isolates extreme, shock-driven instability.
- The moderate threshold captures routine volatility, seasonal fluctuation spillovers, and moderate cyclical variation.

This distinction is central to structural validation. If a model flags numerous deviations in stable years, it may be over-detecting noise rather than isolating true regime breaks.

By comparing strict and moderate thresholds, this study demonstrates that:

1. Extreme structural deviations are temporally concentrated in early 2020.
2. Routine volatility exists across the full time horizon.
3. The strict specification better isolates systemic disruption.

This approach aligns with the structural break literature, which emphasizes the importance of distinguishing persistent regime change from short-term fluctuations (Zeileis et al., 2002). The divergence between strict and moderate thresholds further supports the interpretation that the 2020 disruptions represent statistically exceptional structural shocks rather than ordinary cyclical variation.

Robustness Implications: The sensitivity analysis strengthens confidence in the main findings:

- The clustering of strict anomalies in early 2020 is not an artifact of arbitrary threshold selection.
- Lower thresholds confirm general volatility but do not alter the conclusion that systemic instability was concentrated in the COVID disruption window.
- The model demonstrates controlled detection behavior, avoiding over-identification of breaks.

Thus, the structural propagation narrative remains stable across reasonable parameter adjustments, reinforcing the robustness of the evaluation framework.

6.7. Persistence Analysis via CUSUM

While anomaly detection identifies discrete extreme deviations, cumulative residual analysis evaluates whether shocks accumulate over time. CUSUM therefore provides insight into persistence and regime stability beyond single-period anomalies. To evaluate whether the detected abnormal deviations represent temporary volatility or persistent structural change, cumulative sum (CUSUM) analysis was conducted for each country using the residual series from the baseline model. The CUSUM procedure accumulates residuals over time, allowing identification of regime shifts through sustained directional movements rather than isolated outliers ⁵.

The figure below represents the cross-country CUSUM trajectories.

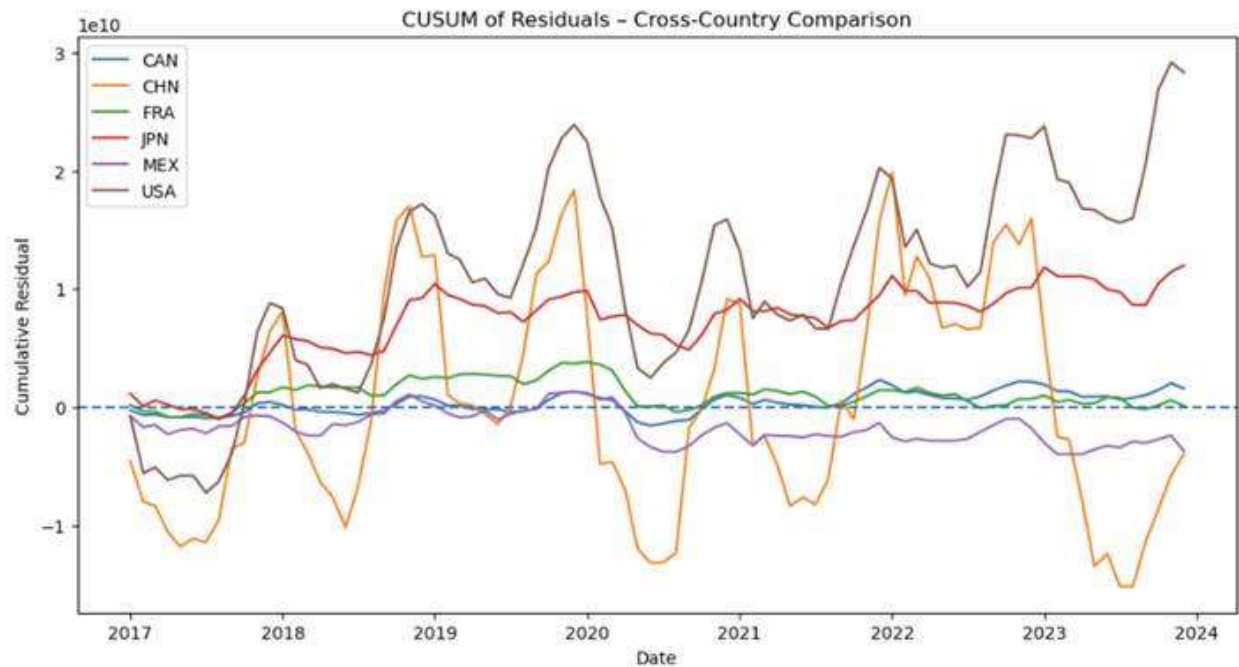


Figure 12: CUSUM of Residuals for Cross-Country Structural Change Detection

Unlike single-period Z-score flags, CUSUM captures cumulative persistence dynamics. A sharp slope change indicates structural instability, whereas a flattening trajectory suggests stabilization.

Structural Disruption Around Early 2020: These slope inflections indicate structural adjustment in the underlying trade process rather than isolated statistical noise. This confirms that the COVID-19 shock was not merely a single-period anomaly but a regime-level disturbance in trade patterns.

In particular:

- China and Japan display pronounced inflection points around early 2020, consistent with the earliest outbreak exposure.
- France and Canada show marked deviations beginning slightly later, aligning with the April transmission phase.
- Mexico exhibits delayed instability consistent with secondary spillover.
- The United States shows a gradual upward drift rather than an abrupt collapse.

The synchronized timing of slope changes across countries strengthens the structural interpretation of the February–May 2020 window identified in the Z-score analysis.

Temporary Shock vs Permanent Regime Shift: Importantly, the post-2021 trajectories for most countries begin to flatten or oscillate around a new equilibrium. This suggests that while the pandemic caused severe short-term instability, the deviations did not accumulate into permanent structural divergence.

This distinction is critical.

A permanent structural break would generate continued divergence in the cumulative residual path. Instead, most countries demonstrate partial reversion or stabilization, indicating adaptive recovery rather than regime replacement.

These findings are consistent with IMF (2021), which notes:

“Trade volumes contracted sharply in early 2020, followed by uneven but observable recovery across regions.”

Similarly, Baldwin and Freeman (2020) argue that pandemic trade disruptions reflected shock transmission through supply networks rather than systemic collapse of the global trading architecture.

The CUSUM trajectories provide cumulative confirmation that the 2020 disruption operated as a high-intensity but temporally bounded structural shock rather than a permanent regime transition. The 2020 shock appears concentrated and intense, yet not permanently regime-altering.

The Case of the United States: Adjustment Without Extreme Z-Score: The United States presents an especially instructive case.

While the strict anomaly threshold ($|Z| \geq 2.5$) did not flag a U.S. structural break in April 2020 ($Z = -2.49$), the CUSUM trajectory reveals a sustained upward drift beginning in 2020. This pattern suggests distributed structural adjustment over time rather than a sharply concentrated one-period collapse.

In contrast to the acute deviations observed in other countries, the U.S. instability manifested as sustained residual accumulation over time rather than a single sharply concentrated collapse month. This distributed adjustment pattern explains why the United States does not appear under the strict anomaly threshold despite exhibiting cumulative structural disturbance.

This helps reconcile why the United States does not appear in the strict anomaly table yet clearly exhibits structural disturbance in cumulative analysis.

Such differences highlight the value of multi-metric validation. Z-scores identify acute shocks. CUSUM captures persistent adjustment dynamics.

Cross-Country Structural Validation: When considered jointly, the CUSUM results:

- Confirm the temporal clustering identified in anomaly detection,
- Demonstrate that the shock produced sustained instability in 2020,
- Show stabilization patterns emerging after 2021,
- Reveal heterogeneity in adjustment mechanisms across countries.

This layered validation approach strengthens the conclusion that the COVID-19 trade disruption was globally synchronized but temporally concentrated and largely transitional rather than permanently regime-altering.

CONCLUSIONS & RECOMMENDATIONS

1. Conclusions

This study examined how international trade patterns change before, during, and after major global disruptions, with a focus on whether the COVID-19 shock represented a temporary structural disturbance or a permanent regime shift. Because traditional trend analysis often struggles to capture abrupt nonlinear volatility during crises, this research developed and validated a residual-based structural detection framework designed to separate true regime shifts from routine cyclical fluctuations.

Using monthly UN Comtrade data for six major economies : China, Japan, Canada, France, Mexico, and the United States. This study combined anomaly detection, cross-country timing comparison, severity interpretation, sensitivity testing, and cumulative residual (CUSUM) diagnostics. The dual implementation in Python and R strengthened reproducibility and reduced the risk of software-specific interpretation.

Under the strict anomaly threshold ($|Z| \geq 2.5$), abnormal deviations clustered tightly between February and May 2020 in a staggered sequence. China and Japan exhibited the earliest structural deviations, followed by Canada and France, with spillover effects in France and Mexico. This phased timing supports the interpretation that disruption propagated through interconnected production networks rather than as a synchronized global collapse.

The United States provides an important nuance. Although it did not exceed the strict anomaly threshold in April 2020, its CUSUM trajectory revealed sustained residual drift beginning in early 2020. This indicates that U.S. instability manifested as distributed structural adjustment over time rather than as a single extreme collapse month. The combination of anomaly detection and cumulative persistence analysis, therefore, provides a more complete structural interpretation than either method alone.

Sensitivity testing further clarified the results. Lowering the threshold to $|Z| \geq 2.0$ substantially increased the number of flagged months across both disruption and non-disruption periods, showing that moderate volatility is common in international trade data. This confirms that the strict threshold more effectively isolates statistically exceptional structural shocks from ordinary cyclical variation.

Additionally, the Arima modeling demonstrated the best results and strongest performance among the evaluated models, as it produced RMSE and MAE values that were substantially lower. This suggests that modeling autocorrelation and temporal dependence is highly critical for predicting and capturing trade dynamics, especially during disruptive periods where there are conditions of structural distortion and volatility.

Overall, the findings suggest that the COVID-19 disruption produced a high-intensity but temporally bounded structural shock. Trade instability was significant, cross-country propagation was staggered, and most economies showed stabilization after 2021 rather than permanent regime replacement.

While the analysis focuses on a single commodity category and a limited set of countries, the framework is designed to be scalable and can be extended to broader datasets and additional sectors. Future research could

expand this approach by incorporating multiple commodities, higher-frequency data, or alternative structural break techniques to further validate and generalize the findings.

More broadly, this study demonstrates how structural diagnostics can complement traditional forecasting methods when analyzing crisis-driven volatility. By emphasizing disruption timing, propagation patterns, and persistence dynamics, the proposed framework provides a practical analytical tool for policymakers and supply-chain analysts seeking to detect and interpret future global trade disruptions.

2. Recommendations

2.1 Policy Recommendations

Policymakers and supply-chain analysts should adopt monitoring systems that integrate both anomaly thresholds with persistence diagnostics. Point-based anomaly detection alone may miss gradual structural adjustment, as observed in the United States, while cumulative diagnostics improve early identification of systemic instability.

The staggered propagation pattern also suggests that resilience planning should account for lagged transmission effects. Disruptions do not occur simultaneously across economies; instead, they diffuse through interconnected trade and supplier networks. Therefore, diversification, upstream risk mapping, and cross-border exposure monitoring remain critical for structural resilience.

2.2 Methodological Recommendations

Future research on disruption-driven volatility should extend beyond predictive accuracy metrics such as RMSE and incorporate structural validation criteria, including timing consistency, persistence analysis, cross-country alignment, and threshold sensitivity testing.

Cross-platform implementation in Python and R should also be encouraged, as replication across analytical environments improves transparency, reduces computational bias, and strengthens empirical credibility.

2.3 Limitations and Future Research

This study is based on aggregate commodity-level trade data, which may mask sector-specific instability patterns. In addition, while threshold sensitivity testing was used to assess robustness, alternative change-point methods may provide additional insight into regime transitions. Future research could apply the same framework to other disruption contexts, including geopolitical conflict, trade policy shocks, and climate-related supply-chain events to evaluate how broadly the framework generalizes beyond the COVID-19 benchmark.

Overall, the proposed framework provides a practical and scalable approach for detecting, validating, and interpreting structural trade disruptions in a rapidly changing global environment.

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